

Frequency Analysis with Index of Coincidence

CS 5158 6058 Project 1

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Course	CS 6058
Operating system	macOS
Programming language	Python (Python 3.14)

This report details how I analyzed letter frequency in English, constructed the Vigenere cipher, performed an Index of Coincidence attack, and ran a batch attack on the ciphertexts in the class shared folder.

Environment Build and Program Use

I developed and ran the project on macOS [insert version]. I used Python [insert the exact output of `python3 --version`]. Python does not compile into C/C++ object files. To follow the required directory structure, my build script packages all source code from `src` into the runnable executable archive `build/ic_project.pyz`.

I installed the dependency and built the program from the project root with:

```
python3 -m pip install -r requirements.txt
python3 build_project.py
```

The build command places the executable only in the build folder. I then use the same command pattern for every function: `python3 build/ic_project.pyz` followed by the function name and its arguments. The data folder contains all inputs and all results generated by the program.

Task 1: English Frequency and Index of Coincidence

I ran the frequency analysis from the project root with this command: `python3`

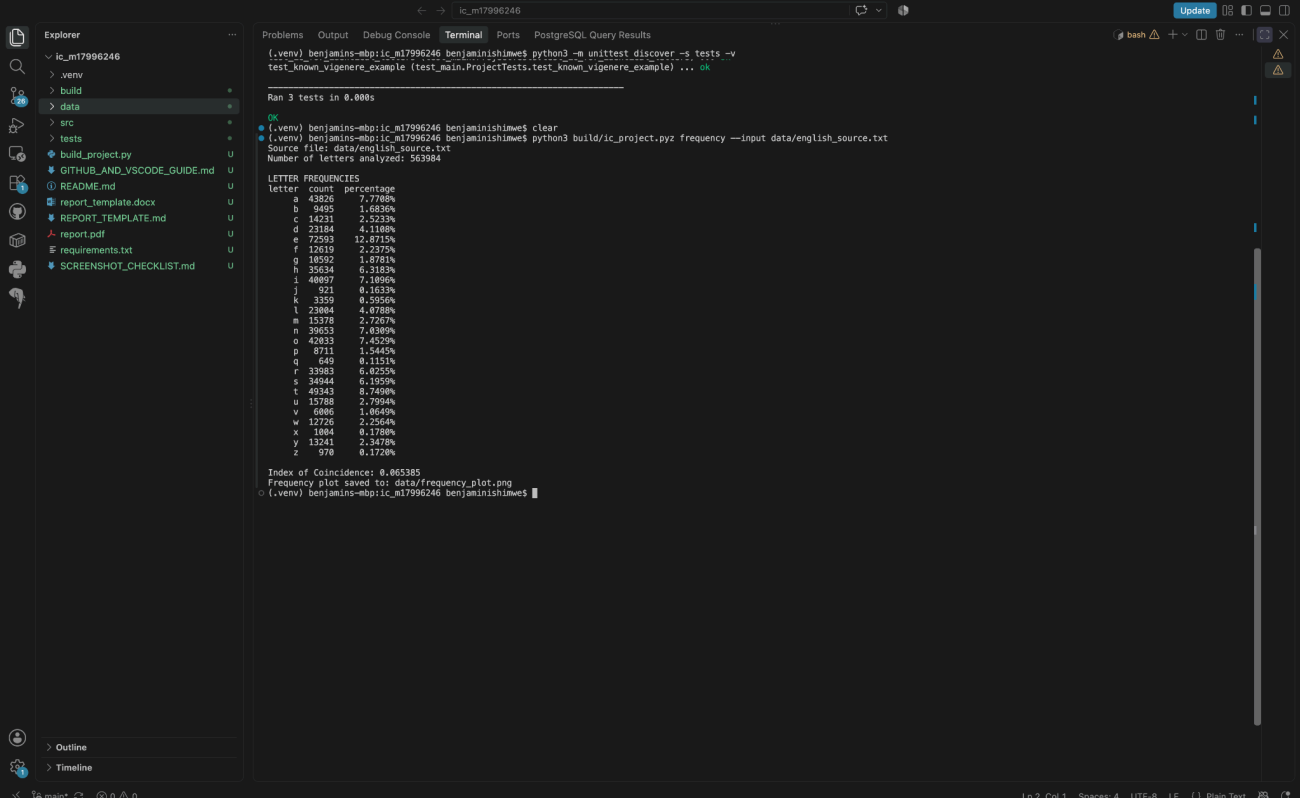
```
build/ic_project.pyz frequency --input data/english_source.txt
```

My text source was *Pride and Prejudice* by Jane Austen, Project Gutenberg eBook 1342. The source is <https://www.gutenberg.org/ebooks/1342>. After removing digits, spaces, punctuation, and other nonalphabetic symbols, the program analyzed 563,984 lowercase English letters. The frequency function counted every letter from a through z.

I calculated IC as the sum of $f_i(f_i - 1)$ for all 26 letters divided by $N(N - 1)$, where f_i is a letter count, and N is the number of retained letters. My result was 0.065385. This is close to

CS 5158 6058 Project 1

the expected English value of about 0.066, so the result is consistent with the value discussed in class. A small difference is expected because letter frequencies depend on the selected text.



```
(.venv) benjamins-mbp:ic_m17996246 benjaminishimw$ python3 -m unittest discover -s tests -v
test_known_vigenere_example (test_main.ProjectTests.test_known_vigenere_example) ... ok

Ran 3 tests in 0.000s

(.venv) benjamins-mbp:ic_m17996246 benjaminishimw$ clear
(.venv) benjamins-mbp:ic_m17996246 benjaminishimw$ python3 build/ic_project.pyz frequency --input data/english_source.txt
Source file: data/english_source.txt
Number of letters analyzed: 563984

LETTER FREQUENCIES
letter count percentage
a 43826 7.7788%
b 9455 1.6836%
c 14231 2.5233%
d 23184 4.1188%
e 72593 12.8715%
f 12619 2.2375%
g 18592 3.2981%
h 35634 6.3183%
i 40897 7.2696%
j 921 0.1633%
k 3359 0.5956%
l 23884 4.2349%
m 15378 2.7267%
n 39653 7.0309%
o 42833 7.6029%
p 8711 1.5445%
q 649 0.1151%
r 33983 6.0255%
s 34944 6.1959%
t 49343 8.7498%
u 15768 2.7944%
v 6886 1.2211%
w 12726 2.2564%
x 1084 0.1923%
y 13241 2.3478%
z 978 0.1720%

Index of Coincidence: 0.065385
Frequency plot saved to: data/frequency_plot.png
(.venv) benjamins-mbp:ic_m17996246 benjaminishimw$
```

Problems Output Debug Console **Terminal** Ports PostgreSQL Query Results

```
● (.venv) benjamins-mbp:ic_m17996246 benjaminishimwe$ python3 build_project.py
Build completed: build/ic_project.pyz
Run it with: python3 build/ic_project.pyz --help
● (.venv) benjamins-mbp:ic_m17996246 benjaminishimwe$ python3 build/ic_project.pyz --help
usage: python3 build/ic_project.pyz [-h]
                                   {frequency,encrypt,decrypt,attack,attack-folder} ...

CS 5158/6058 Project 1 - Index of Coincidence and Vigenere cipher.

positional arguments:
  {frequency,encrypt,decrypt,attack,attack-folder}
    frequency          Task 1: frequency table, plot, and IC
    encrypt            Task 2: Vigenere encryption
    decrypt            Task 2: Vigenere decryption
    attack             Task 3: recover a Vigenere key
    attack-folder      Task 4: attack every .txt file in a folder

options:
  -h, --help            show this help message and exit
● (.venv) benjamins-mbp:ic_m17996246 benjaminishimwe$ python3 -m unittest discover -s tests -v
test_frequency_ignores_nonletters_and_lowercases (test_main.ProjectTests.test_frequency_ignores_nonletters_and_lowercases) ... ok
test_ic_for_identical_letters (test_main.ProjectTests.test_ic_for_identical_letters) ... ok
test_known_vigenere_example (test_main.ProjectTests.test_known_vigenere_example) ... ok

Ran 3 tests in 0.000s

OK
● (.venv) benjamins-mbp:ic_m17996246 benjaminishimwe$ clear
● (.venv) benjamins-mbp:ic_m17996246 benjaminishimwe$ python3 build/ic_project.pyz frequency --input data/english_source.txt
Source file: data/english_source.txt
Number of letters analyzed: 563984

LETTER FREQUENCIES
letter count percentage
a 43826 7.7708%
b 9495 1.6836%
c 14231 2.5233%
d 23184 4.1108%
e 72593 12.8715%
f 12619 2.2375%
g 10592 1.8781%
h 35634 6.3183%
i 40097 7.1096%
j 921 0.1633%
k 3359 0.5956%
l 23004 4.0788%
m 15378 2.7267%
n 39653 7.0309%
o 42033 7.4529%
p 8711 1.5445%
q 649 0.1151%
r 33983 6.0255%
s 34944 6.1959%
t 49343 8.7490%
u 15788 2.7994%
v 6006 1.0649%
w 12726 2.2564%
x 1004 0.1780%
y 13241 2.3478%
z 970 0.1720%

Index of Coincidence: 0.065385
Frequency plot saved to: data/frequency_plot.png
○ (.venv) benjamins-mbp:ic_m17996246 benjaminishimwe$
```



Task 2: Vigenere Encryption and Decryption

I used the key bearcats, which has eight letters. The program retained exactly the first 100 lowercase letters of my input:

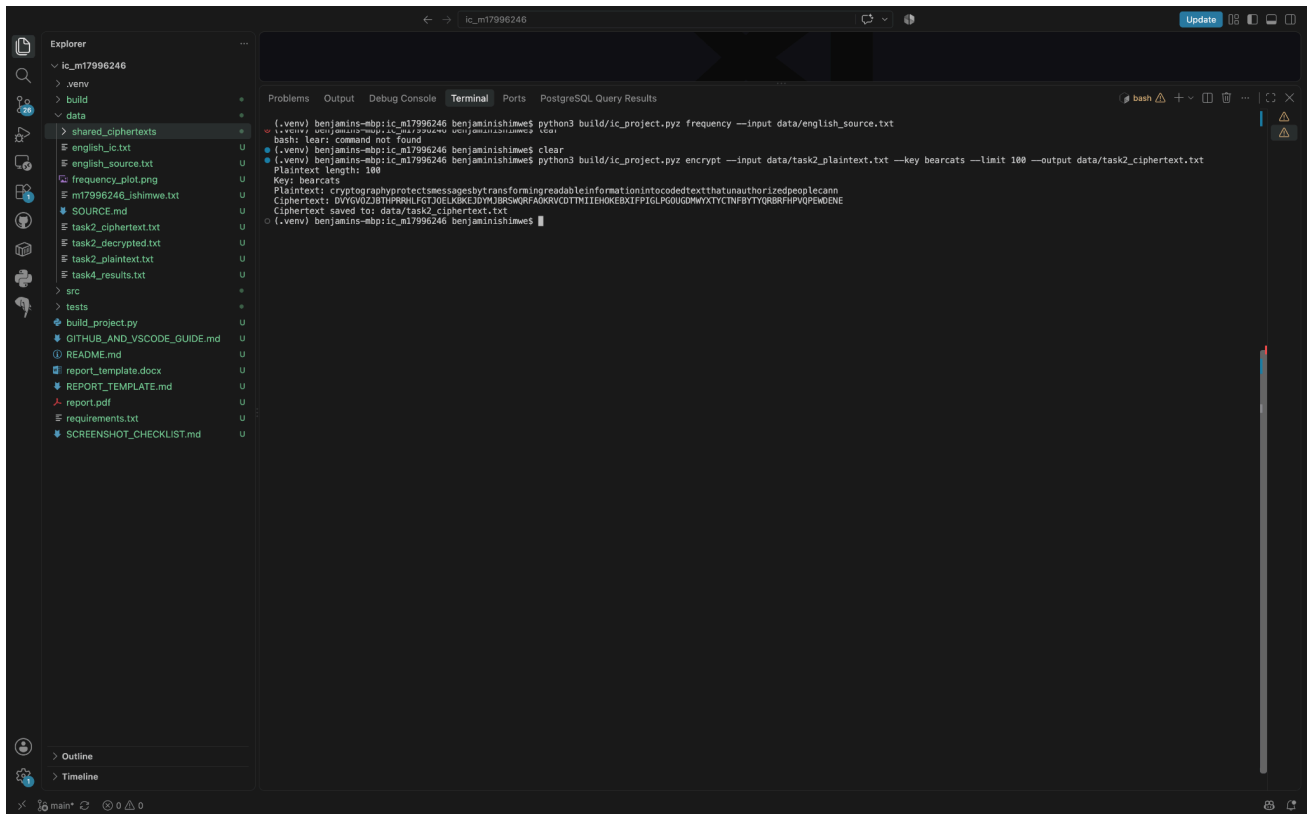
cryptographyprotectsmessagesbytransformingreadableinformationintocodedtextthatunauthorizedpeoplecann

I encrypted the text with:

```
python3 build/ic_project.pyz encrypt --input data/task2_plaintext.txt --key bearcats --limit 100 --output data/task2_ciphertext.txt
```

For each plaintext letter and repeating key letter, encryption adds their values modulo 26. The program produced the following 100 uppercase letters:

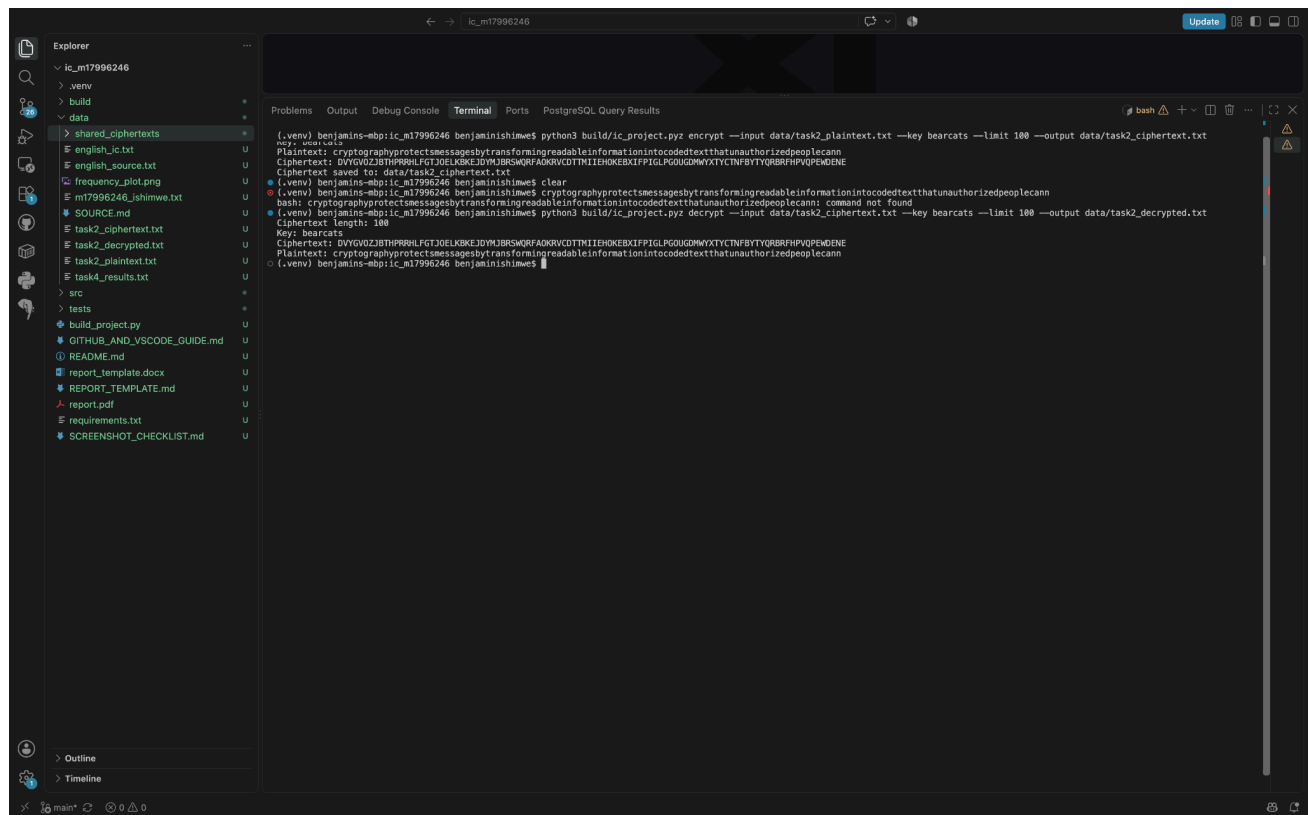
DVYGVQZJBTHPRRHLFGTJOELKBKEJDYMBJRSWQRFQAKRVCDTTMIEHOKEBXIFPIGLPGOUGDMWYXTY
CTNFBYTYQ RBRFHPVQPEWDENE



I decrypted the ciphertext with:

```
python3 build/ic_project.pyz decrypt --input data/task2_ciphertext.txt --key bearcats --limit 100 --output data/task2_decrypted.txt
```

Decryption subtracts each repeating key value modulo 26. The recovered lowercase text matched the original 100 letters exactly.



Task 3: Attack with Index of Coincidence

I generated the long uppercase ciphertext with:

```
python3 build/ic_project.pyz encrypt --input data/english_source.txt --key bearcats --output data/m17996246_ishimwe.txt
```

CS 5158 6058 Project 1

The resulting ciphertext contains 563,984 letters, which is greater than the required minimum of 10,000. I ran the attack with:

```
python3 build/ic_project.pyz attack --input data/m17996246_ishimwe.txt --english-ic data/english_ic.txt
```

For every possible key length from 3 through 19, I separated the ciphertext into columns and calculated the average column IC. The candidate length 8 had average IC 0.065385, which matched the English IC. Length 16 also had the same average IC because it is a multiple of 8, so the program applies a small length penalty and selects the shorter fundamental period.

IC identifies the likely key length but cannot identify a Caesar shift because shifting letters does not change IC. I therefore tested all 26 shifts in each column with chi-square comparison against standard English frequencies. The smallest score in each position produced b, e, a, r, c, a, t, and s. The recovered key length was 8 and the recovered key was bearcats, which exactly matched my key.

```
rypt --input data/task2_ciphertext.txt --key bearcats --limit 100 --output data/task2_decrypted.txt
Ciphertext length: 100
Key: bearcats
Ciphertext: DVYGV0ZJBTHPRRHLFGTJ0ELKBKEJDYMBJRSWQRFADKRVCDTMIIEHOKEBXIFPIGLPGOUGDMWXYTYCTNFBYTYQBRFHPVPQEWEDENE
Plaintext: cryptographyprotectsmessagesbytransformingreadableinformationintocodedtextthatunauthorizedpeoplecann
(.venv) benjamins-mbp:ic_m17996246 benjaminishimwe$ clear
(.venv) benjamins-mbp:ic_m17996246 benjaminishimwe$ python3 build/ic_project.pyz attack --input data/m17996246_ishimwe.txt --english-ic data/english-ic.txt
KEY-LENGTH EVIDENCE (smaller selection score is better)
length average_IC distance_from_English_IC selection_score
3 0.041513 0.023872 0.024922
4 0.052796 0.012589 0.013989
5 0.041513 0.023872 0.025622
6 0.044324 0.021061 0.023161
7 0.041514 0.023871 0.026321
8 0.065385 0.000000 0.002800 <-- selected
9 0.041514 0.023871 0.027021
10 0.044323 0.021062 0.024562
11 0.041513 0.023872 0.027722
12 0.052800 0.012585 0.016785
13 0.041514 0.023871 0.028421
14 0.044321 0.021064 0.025964
15 0.041514 0.023871 0.020121
16 0.065385 0.000000 0.005600
17 0.041513 0.023872 0.029822
18 0.044322 0.021063 0.027363
19 0.041510 0.023875 0.030525

KEY-LETTER EVIDENCE (chi-square; smaller is better)
position 1: b:383.93, h:259628.92, o:298155.09, n:299570.17, m:332246.81 -> b
position 2: e:360.19, k:263839.27, q:286958.52, r:310100.58, p:327448.28 -> e
position 3: a:331.06, g:263787.30, m:282542.33, n:305329.61, l:335390.62 -> a
position 4: r:423.14, x:264218.33, d:293971.98, e:308221.22, c:333728.22 -> r
position 5: c:427.24, i:262582.11, o:301613.65, p:310140.06, n:340194.94 -> c
position 6: a:354.67, g:261956.70, m:294385.77, n:313286.80, l:328704.73 -> a
position 7: t:375.24, z:263652.76, f:284556.17, g:302774.78, e:347467.98 -> t
position 8: s:373.09, y:261236.47, e:296488.34, f:307041.51, d:336890.47 -> s

Recovered key length: 8
Recovered key string: bearcats
(.venv) benjamins-mbp:ic_m17996246 benjaminishimwe$
```

Task 4 : Shared Ciphertext Attack

I uploaded my uppercase ciphertext file, named m17996246_ishimwe.txt, to the class shared folder. It contains 563,984 uppercase letters and uses the key bearcats. I then downloaded every class .txt file into data/shared_ciphertexts and ran:

```
python3 build/ic_project.pyz attack-folder --folder data/shared_ciphertexts -- english-ic data/english_ic.txt
```

The program generated data/task4_results.txt in the required format:

```
data > E:\task4_results.txt
1 filename: m14257834_khalil.txt recovered key string: seven
2 filename: m14618318_lohi.txt recovered key string: sunrise
3 filename: m14971350_kunare.txt recovered key string: thatskyelsthegast
4 filename: m14998897_shapiro.txt recovered key string: idonthow
5 filename: m15318861_alrefai.txt recovered key string: bearcat
6 filename: m15538363_devils.txt recovered key string: cincinnati
7 filename: m16633275_lee.txt recovered key string: rabbitisgood
8 filename: m16638755_andrew.txt recovered key string: yourkey
9 filename: m17309892_cramer.txt recovered key string: bemoovie
10 filename: m17996246_ishimwe.txt recovered key string: bearcats
11 filename: m18894882_cho.txt recovered key string: chosecurity
12
```

```
(.venv) benjamin@mbp:ic_m17996246$ python3 --help
Build completed: build/ic_project.pyz
Run it with: python3 build/ic_project.pyz --help
(.venv) benjamin@mbp:ic_m17996246$ python3 build/ic_project.pyz attack-folder --folder data/shared_ciphertexts --english-ic data/english_ic.txt
TASK 4 SUCCESS
filename: m14257834_khalil.txt recovered key string: seven
filename: m14618318_lohi.txt recovered key string: sunrise
filename: m14971350_kunare.txt recovered key string: thatskyelsthegast
filename: m14998897_shapiro.txt recovered key string: idonthow
filename: m15318861_alrefai.txt recovered key string: bearcat
filename: m15538363_devils.txt recovered key string: cincinnati
filename: m16633275_lee.txt recovered key string: rabbitisgood
filename: m16638755_andrew.txt recovered key string: yourkey
filename: m17309892_cramer.txt recovered key string: bemoovie
filename: m17996246_ishimwe.txt recovered key string: bearcats
filename: m18894882_cho.txt recovered key string: chosecurity
Summary saved to: data/task4_results.txt
(.venv) benjamin@mbp:ic_m17996246$ find data/shared_ciphertexts -maxdepth 1 -type f -name "*.txt" | wc -l
11
(.venv) benjamin@mbp:ic_m17996246$ find data/task4_results.txt
11
(.venv) benjamin@mbp:ic_m17996246$ find data/shared_ciphertexts -maxdepth 1 -type f -name "*.txt" | wc -l
11
(.venv) benjamin@mbp:ic_m17996246$ find data/task4_results.txt
11
(.venv) benjamin@mbp:ic_m17996246$
```